

ELECTIVE COURSE-5.5

URA752: COMPUTER VISION AND AUGMENTED REALITY				
	L	T	P	Cr
	2	0	2	3
Course Objective: To become familiar with the concept and applications of computer vision and augmented & reality and learn different types of algorithmic techniques and strategies.				
Syllabus Introduction of Augmented Reality (AR): Definition and Applications, History, Types of AR, Suitable devices, Virtual and Mixed reality, Holograms, AR Displays, Method of Augmentation, Spatial Display Model, Occlusion. Tracking in AR: Basic steps of AR, Tracking, Calibration, Registration, Co-ordinate Systems: Model-View-Projective Transformation, Frame of reference, Characteristics of Tracking Technology: Physical Phenomenon, Triangulation, Trilateration, Measurement Principles, Degree of Freedom, Stationary Tracking System, Mobile Tracking, Optical Tracking, Sensor Fusion. Fundamentals of Computer Vision: Definition and applications, Image formation, Image enhancement, Histogram processing, Edge detection, Image segmentation, Clustering Types of noise, Image restoration, Video frames, Key frames, Background subtraction, Moving object detection. Computer Vision in AR: Computer Vision for Marker Tracking, Thresholding, Contour detection, Hough Transformation, Quadrilateral fitting, Feature extraction: SIFT, LBP, Pose Estimation, Homography, Incremental Tracking, SLAM: Bundle Adjustment, Parallel Tracking and Mapping, Outdoor Tracking, STML.				
Laboratory Work To implement various techniques studied during course.				
Course Learning Outcomes (CLOs) The students will be able to: 1. analyze the components of AR systems, its current and upcoming trends, types, platforms, and devices. 2. understand the basic steps and technologies required to achieve AR system. 3. understand the fundamental concepts of computer vision and image/video processing. 4. demonstrate the building and integration of various well-known computer vision algorithms in the AR.				
Text Books 1. Dieter Schmalstieg, Tobias Höllerer, Augmented-Reality-Principles-and-Practice-Usability- , Addison-Wesley (2016) 1st ed. 2. Szeliski, R., Computer Vision: Algorithms and Applications, Springer-Verlag London Limited (2011), 1st Edition.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Reference Books

1. Aukstakalnis S., Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR, Addison-Wesley (2016).
2. Gonzalez, C., R. and Woods, E., R. Digital Image Processing, Addison- Wesley (2018) 4th Edition

Evaluation Scheme:

S N	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessional (Assignments/Projects/Tutorials/Quizzes)	35

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.